

浙江大学



Lab Practice Report for Undergraduate

Name:

College: ZheJiang University

Department: Science

Major: Marine Science

Student NO.:

Instructor: Yanan Di

Date: 2024.11.14

Lab Practice Report of Zhejiang University

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Course Name: Marine biology Type of Practice: dissection

Name of Practice: Anatomy of Typical Mollusca-Cephalopoda

Student Name: _____ Major: Marine biology

Student Number: _____

Team Member: Zhiang Liu

Instructor: Yanan Di

Practice Location: 241 Practice Date: 2024.11.14

1. Aims and Requirement of the Practice (Compulsory)

1. Learn about Cephalopoda biology and complete a live or a virtual dissection of cuttlefish.
2. Identify the external anatomy of cuttlefish; describe the function of important external features.
3. Identify the major internal organs of cuttlefish; describe their biological functions.
4. Demonstrate dissection skills

2. Background information of the Practice

As the most morphologically and behavioral type in the phylum mollusk, cephalopoda has the most complex brain among invertebrates. Cephalopods are characterized by a completely merge head and foot, with a ring of arms and/or tentacles.

Nautiloidea have more than 90 tentacles, members of orders Sepioidea and Teuthoidea have eight arms and two tentacles. Ocotopoidea and Vampyromorpha have eight arms.

The mantle surrounds the visceral sac and possesses strong muscles required for contraction of the cavity and respiration. An opening in the mantle cavity serves as an inhalant aperture, whereas the funnel serves as the exhalant aperture.

An external shell is possessed only by the Nautiloidea. Sepioidea and Teuthoidea have reduced inner shells, while Octopoda and Vampyromorpha lack shells altogether. A cephalopod is also characterized by a horny beak secreted by the walls of the buccal cavity, and a radula within the buccal cavity.

It also has a complete nervous system and sensory organs like a large brain and big eyes.

3. Equipment and materials

1. organisms: cuttlefish
2. Dissection set
3. Ice box and debris ice
4. 1.5ml centrifuge tubes and racks
5. Liquid nitrogen and container
6. Marker pens
7. Gloves, plastic bags, paper towels.

4. Procedures

1. Select one organism and put it on ice. Observe it to identify its gender and basic body structure. Measure the size and make record.

2. Dissection of cuttlefish

3. Homogenization and Preservation of tissue samples

- (1) Cut muscle sample and digest sample (no bigger than 0.27 cm^3 , 3 replicates)

- (2) Put 1.5ml centrifuge tubes, add sample into it. Add 500 μ l PBS into each tube. Homogenize it 1min with homogenizer 1min. Then add 500 μ l PBS again and homogenization 10s.

- (3) Put tubes into centrifuge, spin at 4°C, 12000g for 5min. When it finishes, collect supernatant and transport it to another clear 1.5ml tube. Mark it and send it to -20°C Refrigerator

- (4) Clean up everything you used. Fold all materials into paper towels and put into plastic bags. Rinse and sanitize dissection equipment.

5. Questions

- (1) Take pictures of your samples identify the external structures and functions

SIZE (long x wide x high): 29.81cm x 3.91 cm x 1.88 cm



Tentacles and arms	Capture foods
Fin	Swim
Water jet	Assisted movement
eye	Observe
mantle	protects internal organs

- (2) Dissect the organisms and make forms to show different internal structures of cuttlefish



Ink sac:



gills:



Heart:



(3) Discuss the internal structures and their functions.

structures	functions
Stomach	Digestive sites
Heart	Blood supply
intestine	Absorb nutrients from food
Gills	Help breathe
salivary glands	Secrete digestive enzymes
mantle	protects internal organs
Funnel and siphon	drain
Ink sac	Used to produce ink
digestive gland	Secrete digestive enzymes

(4) What is the most impressive body structure of Cephalopoda during your dissection, and explain the reason

What impressed me the most was the structure of the digestive glands, because the yellow contents were filled with it, which made me feel a little sick to my stomach

(5) Sample quality table

Sample number	tube quality(g)	Sample and tube quality (g)	Sample quality (g)
M1	0.8332	0.9879	0.1547
M2	0.8469	0.9968	0.1499
M3	0.8272	0.9757	0.1485
D1	0.8343	0.9928	0.1585
D2	0.8207	0.9654	0.1447
D3	0.8359	0.9924	0.1565

注：不同类型的实验课对实验报告可有不同要求，各个学科的实验报告可以根据自己的学科特点做适当的调整，但上述基本内容中的第一、二、三、六条为必须填写的内容。

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